BTS–VIII(R)–04–23–1581 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

C

B.Tech. Degree VIII Semester Regular Examination April 2023

CS 19-202-0808 BLOCKCHAIN

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand how block chain systems work.
CO2: Familarize the functional / operational aspects of cryptocurrency ecosystem.
CO3: Design, build and deploy smart contract and distributed applications.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Illustrate CAP theorem.	3	L1	1	1
	(b)	Discuss about consensus mechanisms.	3	L2	1	1
	(c)	What is bitcoin?	3	L1	2	1
	(d)	Explain UTXO and coin base transactions.	3	L1	2	1
	(e)	Write the use of Merkle trees in block chain.	3	L1	2	1,2
	(f)	Write short notes on history of cryptocurrency.	3	L1	2	1
	(g)	Explain etherium.	3	L1	3	3
	(h)	Discuss Hyperledger Fabric.	3	L2	4	2,3

PART B

(4 × 12 = 48)

		Marks	BL	CO	PO
II.	Discuss the different types of block chains.	12	L1	1	1
	OR				
III.	List the comparison between centralized and decentralized systems (networks/applications) and write notes on routes to decentralization.	12	L2	1	2
IV.	Discuss about address and keys in bitcoin.	12	L1	2	1
	OR				
V.	Discuss about transactions in block chain. Also explain transaction structure and its types.	12	L1	2	1
VI.	Briefly explain the components of Ethreum ecosystem.	12	L2	3	2
	OR				
VII.	Write short notes on: (i) Truffle (ii) Solidity language.	12	L1	3	1
VIII.	Briefly explain hyperledger.	12	L2	4	1
	OR				
IX.	Explain how blockchain can be used for any supply chain application.	12	L2	4	2

Blooms's Taxonomy Levels

L1 – 62.5%, L2 – 37.5%.

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BTS-VIII(R/S)-04-24-2302 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

C

B.Tech. Degree VIII Semester Regular/Supplementary Examination April 2024

CS 19-202-0808 BLOCKCHAIN
(2019 *Scheme*)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Understand how block chain systems work.
CO2: Familarize the functional / operational aspects of cryptocurrency ecosystem.
CO3: Design, build and deploy smart contract and distributed applications.
CO4: Identify a use case for a blockchain application.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

			Marks	BL	CO	PO
I.	(a)	Discuss the concept of double spending. (8 × 3 = 24)	3	L1	1	1
	(b)	Discuss about genesis block and UTXO.	3	L2	1	1
	(c)	What is proof of work? What are its alternatives?	3	L1	2	1
	(d)	Explain Brewer's theorem.	3	L1	2	1
	(e)	What is Remix and its role in block chain?	3	L1	2	1,2
	(f)	Write short notes on mining node and EVM node.	3	L1	2	1
	(g)	Explain the concept of mining.	3	L1	3	3
	(h)	Discuss about smart contracts.	3	L2	4	2,3

PART B
(4 × 12 = 48)

		Marks	BL	CO	PO
II.	What is a blockchain? Explain the different types of blockchain.	12	L1	1	1
	OR				
III.	List the comparison between centralized and decentralized systems (networks/applications) and distributed systems.	12	L2	1	2
IV.	Illustrate asymmetric cryptography, public and private keys in the context bitcoin.	12	L1	2	1
	OR				
V.	Explain how the data in a block is validated and how security is ensured in a block chain.	12	L1	2	1

(P.T.O.)

		Marks	BL	CO	PO
VI.	What are the componants of ethreum eco-system? Discuss any two tools used in the development of the eco-system.	12	L2	3	2
	OR				
VII.	What are the functions modifiers in solidity? Mention the most widely used modifiers.	12	L1	3	1
VIII.	Define Hyperledger and draw the reference architecture services of hyperledger and explain each component.	12	L2	4	1
	OR				
IX.	What are the limitations of block chain? Explain in detail.	12	L2	4	2

Bloom's Taxonomy Levels
L1 – 62.5%, L2 – 37.5%.

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